Gabe Brazil

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Education

University of Wisconsin-Madison Madison, WI
B.S. Computer Science & B.S. Economics (Math Emphasis) *Expected Graduation: May 2028*
Certificate: Data Science

GPA: 3.53

Honors: Eagle Scout

Experience

Research Intern, Neuroscience Lab (Prof. Rizvi) June 2024 – August 2024
University of Wisconsin-Madison *Madison, WI*
- Collaborated with postdocs to program FPGA microchips for calcium detection experiments on mice brains.
- Learned Vivado and contributed to neuroscience hardware/software research.

Automation Engineering Intern June 2024
MeiraGTx *London, UK*
- Programmed Hamilton STAR liquid-handling robot using proprietary VENUS system.
- Created pipetting methods to test lab chemicals and ran a serial dilution experiment for drug development.

Eagle Scout Project Leader – Music Room Revitalization June 2024
Scouting America *New York, NY*
- Crowdfunded $2500 and led a group of 10 volunteers to transform a school music room (Atlas Foundation for Autism) by repainting the entire space and building new furniture.
- Created a well-edited tutorial video on how to use FL Studio digital music software for Students with Autism.

Projects

Arbitrage Trader Script | *Python, PyTorch, React, Vite, SQLite*
- Built a full-stack arbitrage scanner that pulls live odds from betting and prediction market APIs (Kalshi, Polymarket, and others) to detect cross-platform pricing discrepancies (arbitrage).
- Trained a small sentence embedding model with PyTorch to match event listings across markets with inconsistent naming, ensuring multiple listings reference the same actual event.
- Logged completed transactions to an SQLite storage layer and built a React/Vite frontend to analyze trade performance.

Language-Learning AI App | *HTML, Python*
- Developed a full-stack, AI-powered language-learning application during a 24-hour Hackathon (MadHacks 2025).
- Worked seamlessly with three teammates to deliver a functional prototype on deadline.
- Integrated third-party speech-to-text and text-to-speech AI models into the backend using vendor documentation.

Peer-to-Peer Webcam/Chatroom Game | *React, Express.js, Socket.IO, WebRTC*
- Built a real-time video chat platform with peer-to-peer webcam streaming in private/public rooms.
- Implemented gameplay where users choose one of four random YouTube videos to share with peers.
- Developed backend signaling server for room management, chat, and video session handling.
- Integrated Discord OAuth with JWT cookies for secure authentication.

Technical Skills

Web & Backend: React.js, Node.js, Express.js, Socket.IO, WebRTC, JWT, Cookies/CORS
Languages: JavaScript, Python, Java, SQL
Tools & Platforms: MySQL Workbench, Railway, Jupyter Notebook, Postman, Agentic AI Integration
Machine Learning: PyTorch, NumPy, Pandas, scikit-learn